

Postdoctoral Application File

Research Portfolio & Curriculum Vitae

Ferdaous MASMOUNI

PhD in Electrical Engineering
Energy Systems Modeling, Control & Optimization



Sfax, Tunisia · French residence permit

Contact

Email: ferdaous.masmoudi@gmail.com

Phone: +216 23 605 813

Address (Sfax, Tunisia): Route Gremda, Sidi Abbes, 3062

Website: ferdaousmasmoudi.github.io

Profiles:

ORCID: 0000-0001-8457-7162

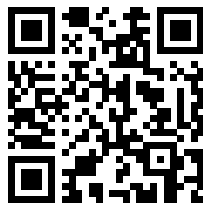
ResearchGate: Ferdaous-Masmoudi

LinkedIn: ferdaous-masmoudi-2a95b64a

GitHub: ferdaousmasmoudi

Online Research Portfolio

Scan to access the full thesis, publications, and research materials.



<https://ferdaousmasmoudi.github.io/>

Ferdaous Masmoudi

PhD in Electrical Engineering
Energy Systems Modeling,
Control & Optimization

Route Gremda
3062 Sfax, Tunisia
French residence permit
☎ +216 23 605 813

✉ ferdaous.masmoudi@gmail.com
🌐 ferdaousmasmoudi.github.io
ID 0000-0001-8457-7162
R[®] Ferdaous-Masmoudi
in ferdaous-masmoudi-2a95b64a
🔗 ferdaousmasmoudi



Research Profile

PhD in Electrical Engineering and Renewable Energy Systems with experience in the end-to-end development of intelligent energy platforms, covering nonlinear mathematical modeling, parameter identification, optimization strategies, embedded implementation, and experimental validation.

Research activities include multi-level mathematical modeling of photovoltaic and electrochemical energy systems, data-driven parameter identification (least squares, ARX-based approaches), and optimization-based tuning under real environmental conditions. State estimation techniques, including Extended Kalman Filtering (EKF), were implemented for real-time monitoring and control-oriented system supervision.

Strong experience in the integration of sensors, real-time data acquisition chains, signal processing, and control algorithms within embedded architectures, complemented by MATLAB/Simulink-based simulation, power electronics implementation, and cloud-enabled supervision platforms for real-world system deployment.

Education

- 2016 **PhD in Electrical Engineering**, *National Engineering School of Sfax (ENIS), University of Sfax, Tunisia*, Thesis: *Optimization of Standalone Photovoltaic Systems*.
- 2012 **MSc in Electrical Conversion and Renewable Energy (CEER)**, *ENIS, University of Sfax, Tunisia*, Research: *Modeling and MPPT Control of a PV Conversion System*.
- 2011 **Engineering Degree in Electrical Engineering (Electro-Technics)**, *ENIS, Sfax, Tunisia*
- 2008 **Preparatory Cycle for Engineering Studies (Mathematics–Physics)**, *IPEIM El Manar, Tunis, Tunisia*
- 2006 **Baccalaureate in Mathematics, with Honors**, *Lycée Pilote Ariana (LPA), Ariana, Tunisia*

Doctoral Research Contributions

Nonlinear Modeling and Identification	Development of nonlinear mathematical models of photovoltaic generators (single-, double-, and multi-diode configurations, panel-level modeling) and electrochemical storage systems, including analytical formulation, state-space representation, and data-driven parameter identification and estimation under real operating conditions.
Optimization and Control	Formulation and implementation of optimization-based control strategies for photovoltaic energy systems, including MPPT algorithms, dual-axis solar tracking mechanisms, and boost-converter power regulation, with performance validation under dynamic irradiance and temperature variations.
Embedded Implementation and Experimental Validation	MATLAB/Simulink-based simulation and model validation, followed by implementation of embedded control architectures (C++/Python) on SoC and microcontroller platforms for real-time operation. Design and realization of a boost converter power stage and dual-axis solar tracking mechanism with motor drivers integration. Development of real-time multi-sensor acquisition systems (voltage, current, irradiance, temperature, GPS, RTC, IMU) including signal conditioning, filtering, and experimental dataset validation. Deployment of cloud-enabled supervision and server-side data logging for remote monitoring and system performance analysis.

Applied Research & Engineering Experience

- 2016–Present **Co-founder and R&D Lead, NOVEL-TI, Sfax, Tunisia**
- Directed applied research and system development in intelligent energy systems, embedded architectures, and AI-enabled IoT platforms.
 - Designed and supervised end-to-end development of hardware–software systems, including mobile robotic platforms, photovoltaic-powered systems, multi-sensor acquisition architectures, and cloud-connected supervision dashboards.
 - Led development of AgriTech solutions integrating LoRa/WiFi connectivity, RFID/NFC logging, and blockchain-based traceability infrastructures (Hyperledger Fabric) for compliance-driven industrial environments.
 - Managed multidisciplinary technical teams and coordinated international collaborations with European industrial partners.

Academic Internships & Early Industrial Training

- 2013–2014 **Research Internship – Battery Observation and Management System (BOMS), LSIS Laboratory, Polytech Marseille, Marseille, France**
- Modeling and embedded implementation of battery monitoring and energy management algorithms, with laboratory validation using MATLAB and microcontroller-based architectures.
- 2012 **Research Internship – Residential PV Integration, TU Chemnitz, Chemnitz, Germany**
- Simulation and component-level optimization of photovoltaic system integration in residential environments.
- 2009–2011 **Industrial Training – Energy and Power Systems (Tunisia):** STEG (National Dispatching), OACA – Carthage Airport, SNDP AGIL. Exposure to grid operation, medium-voltage system design, switchgear manufacturing, and grid-connected photovoltaic system sizing.

Teaching Experience

- 2016–Present Contract-based Lecturer and Research Collaborator. Delivered lectures and laboratory sessions in instrumentation, mechatronics, embedded systems, and open-source development across multiple higher education institutions.

Teaching Institutions: Faculty of Sciences of Sfax (FSS); Higher Institute of Industrial Management (ISGI); Private Institute of Applied Sciences (IPSAS).

Courses and Labs: Computer-Based Instrumentation; Open-Source Software Development; Maintenance of Mechatronic Systems; Mechatronics.

Participation in IEEE SSD'18 and technical workshops on IoT, STM32, Arduino, and smart systems.

Technical Skills

Programming & Build Systems	C++, Python, MATLAB, Bash scripting, GNU Make	Embedded Platforms	STM32, ESP32, Arduino, Jetson, Raspberry Pi, Odroid, Motor Driver
Simulation & Instrumentation	MATLAB/Simulink, LabVIEW, Proteus (ISIS/ARES)	PCB Design& Hardware	Altium, KiCad, EasyEDA, firmware development, bootloader configuration
Communication & IoT	MQTT, LoRa, Zigbee, Wi-Fi, BLE, GSM, Modbus, I2C, SPI, UART	Robotics Middleware	ROS/ROS2, Gazebo, RViz
Mobile & Web Development	Android (Kotlin), REST APIs, Firebase, QR/Camera integration	Cloud & Backend	Real-time data logging, server-side supervision, cloud synchronization
AI & Vision	OpenCV (C++/Python), YOLO, TensorFlow Lite, ONNX, RL(basic)	Platforms & Tools	Linux, Windows, Git

Languages

Arabic (Native) · French (Fluent) · English (Professional proficiency) · German (Basic knowledge)

Selected Publications

- Peer-Reviewed Journals
- Masmoudi, F., "Identification of internal parameters of a mono-crystalline photovoltaic cell model with experimental validation," *International Journal of Renewable Energy Research (IJRER)*, 2014 (Indexed in Scopus).
 - Jarraya, I., Masmoudi, F., et al., "An online state-of-charge estimation for lithium-ion and supercapacitor in hybrid electric drive vehicle," *Journal of Energy Storage*, 2019 (Elsevier, Scopus indexed).
 - Loukil, J., Masmoudi, F., Derbel, N., "A real-time estimator for model parameters and state-of-charge of lead-acid batteries in photovoltaic applications," *Journal of Energy Storage*, 2021 (Elsevier, Scopus indexed).
- International Conferences (IEEE Xplore Indexed)
- SSD 2019 – MPPT control strategies for photovoltaic applications: algorithms and comparative analysis.
 - SSD 2018 – "Modeling and parameters estimation for lithium-ion cells in electric drive vehicle,"
 - SSD 2018 – "Performance analysis of Luo-converter for photovoltaic application,"
 - SSD 2017 – "Design and comparison of quadratic boost and double cascade boost converters with boost converter,"
 - SSD 2017 – "Comparative study of fundamental variable input converters used for PV conversion systems,"
 - SSD 2017 – "State-of-charge estimation of lead-acid battery using a Kalman filter,"
 - SM2C 2017 – "Comparative study of the performances of the DC/DC Luo-converter in photovoltaic applications,"
 - IREC 2016 – "Modeling and simulation of conventional DC–DC converters dedicated to photovoltaic applications,"
 - SSD 2016 – "Single and double diode models for conventional mono-crystalline solar cell with extraction of internal parameters,"
 - SSD 2016 – "Modeling of internal parameters of a lead-acid battery with experimental validation,"
 - STA 2015 – "Design and realisation of a photovoltaic system controlled by an MPPT algorithm,"
 - SAAEI 2014 – "Simple models design for one mono-crystalline photovoltaic cell by identification of internal parameters." Maroc (Tanger).
 - EVER 2012 – "Oil service station PV-based power supply: a challenging Tunisian national program," (Monaco).
- Book Chapters (Springer)
- Loukil, J., Masmoudi, F., Derbel, N., "Third order model and identification of lead-acid batteries using meta-heuristic algorithms and experimental measurements," Book Chapter, 2020.
 - Boujelben, N., Masmoudi, F., Djemel, M., Derbel, N., "Modeling and comparison of boost converter with cascaded boost converters," *Green Energy and Technology*, Book Chapter, 2019.

Online Research Portfolio

Scan to access the full thesis, publications, and research materials.

